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To: Research Council of Finland (RCF)
Project Title: Modelling Turbulent and Magnetised Plasmas at Exascale with AI-assisted Simulation Tools (MomEnt)
Principal Investigator (PI): Prof. Maarit Korpi-Lagg
Call: Special Funding for High-Performance Computing (HPC)

Collaboration Letter

Dear Evaluation Panel,

I am pleased to confirm my collaboration on the *Modelling Turbulent and Magnetised Plasmas at Exascale with AI-assisted Simulation Tools (MomEnt)* project, led by Prof. Maarit Korpi-Lagg. This proposal, submitted to the Research Council of Finland's Special Call for High-Performance Computing, represents a bold and timely effort to push the boundaries of computational astrophysics.

The project addresses one of the most complex challenges in solar and plasma physics: understanding the magnetic dynamo processes of the Sun through the utilization of large-scale, GPU-accelerated simulations. By employing exascale computing, we aim to develop a new generation of AI-powered Plasma simulation that brings new insights into solar magnetism, an understanding essential for predicting Space Weather hazards affecting the Earth. Numerous critical infrastructures, such as satellites, electrical power grids, and communication systems (e.g., navigation via the Global Positioning System; radio communications), remain susceptible to these hazards. Furthermore, in addition to the creation and behavior of plasma observed in natural settings, artificially generated plasmas in laboratories possess a diverse array of technological applications. In particular, Plasma research is crucial in fusion generators where magnetic effects need to be understood to enable the successful production of energy.

Beyond raw computational performance, this project addresses critical challenges in managing petabyte-scale data streams, optimizing AI training workflows, and maintaining a simulation-analysis pipeline that is both high-throughput and interpretable. My group is committed to advancing these areas, aiming to convert computational complexity into new scientific capabilities that align with international priorities in AI, high-performance computing, and solar physics.

I am an Assistant Professor at the University of Texas at Austin and serve as co-PI and AI Lead at the NSF-Simons AI Institute for Cosmic Origins (CosmicAI), established in October 2024. CosmicAI develops advanced AI methodologies to investigate fundamental questions about the origins of the universe. It brings together a consortium of leading institutions, including UT Austin, the University of Utah, the University of Virginia, the National Radio Astronomy Observatory (NRAO), NSF's NOIRLab, Stanford University, and UCLA. The institute's research centers on four core AI themes: trustworthiness, efficiency, interpretability, and robustness. Additionally, it fosters collaboration through open-source tool development, shared data resources, and AI education initiatives aimed at broadening AI accessibility in science communities.

My contributions to this project focus mainly on scalable machine learning, with emphasis on physics-informed neural networks and real-time simulation analysis. In collaboration with Prof. Korpi-Laggs group, we aim to develop AI-driven techniques for efficient interpretation of large-scale solar DNS datasets. A primary goal is the development of robust subgrid-scale models that maintain physical fidelity while reducing computational demands, which is critical for sustainable high-performance computing. Our collaboration will include biweekly virtual meetings, joint development of machine learningaugmented simulation tools, and co-authored publications. Subject to feasibility and funding, we also plan to coordinate research visits and workshops to deepen collaboration. Please note that my participation in this effort does not entail any request for financial support from the RCF.

I support this proposal and look forward to working with Prof. Korpi-Lagg and the Aalto University HPC Lab. Jointly, we have the potential to transform simulations that can potentially revolutionize our understanding of the Sun and push the boundaries of computational astrophysics.

Sincerely,

Prof. Arya Farahi
Assistant Professor
Co-PI, CosmicAI Institute
AI Lead, Explainable Universe